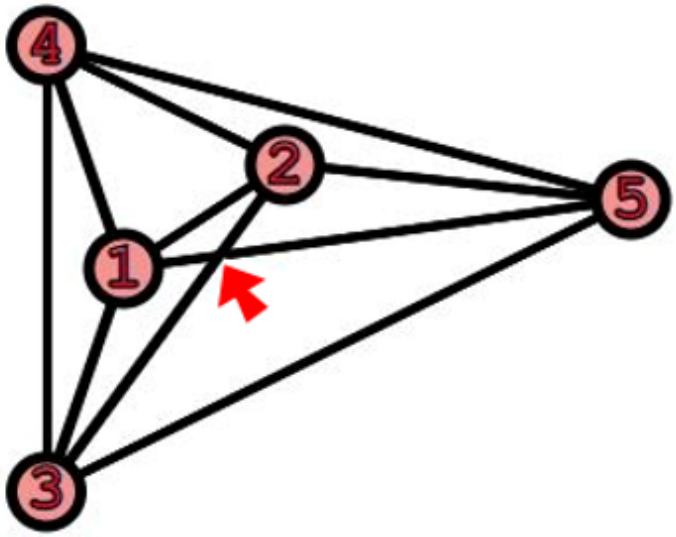


Figure 15.7 A better embedding for K_5. In this case we have only one crossing point, pointed at by the fat arrow. Notice that there are still infinitely many embeddings equivalent to this one…

Therefore, we can say that the answer to this question is no, they aren't all equivalent; as a matter of fact, working to find the "best"[6] possible embedding for graphs will be our quest for the rest of this book[7].

15.2.1 Using Kuratowski's Theorem in Practice

Kuratowski's theorem states that K_5 and $K_{3,3}$ are the "simplest" graphs that don't have a planar embedding. How did he mean "simplest"? Well, in this case the sense is that there isn't any smaller graph (meaning with fewer vertices or edges) that isn't planar, and so every sub-graph of either K_5 or $K_{3,3}$ has a planar embedding.